

TrakRF asset-egress demo — Tim's card

DRAFT for Mike + Tim review. Operator card — physical connections and the app only. No commands, no logins to the box. Anything not on this card = **call Mike**.

The pitch in one line: a tagged piece of equipment leaving through the door sets off the strobe — and random tagged stuff walking by does *not*.

You only ever touch two things: the **physical gear** (plug it in, power it on) and the **app in your browser** (`https://app.demo.trakrf.id`). That's it.

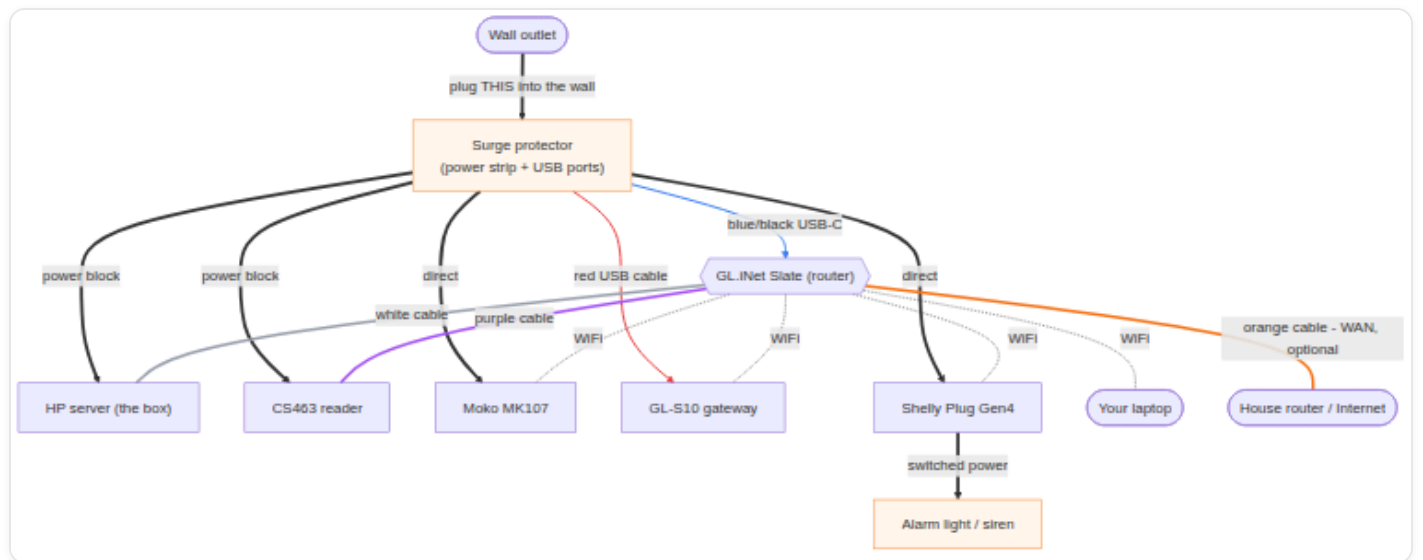
WiFi: network `TrakRF` · password `trakrf.id`

A. Set up the gear — it ships pre-wired

The kit shipped with **everything already plugged into the surge protector** and all the network cables connected. Your normal setup is tiny:

1. Set the pieces roughly in place — the **2–3 antennas at the doorway** (each aimed *across* the opening, not down the hall), the strobe near the door.
2. **Plug the surge protector into a wall outlet.** Everything powers on together.
3. (*Optional — only if the venue has wired internet*) plug the **orange** cable into the house router. The demo works fine without it.
4. Wait **~2 minutes** for the box and reader to boot. **Don't unplug anything.**

Here's how it's all connected, in case a cable comes loose or you need to repack:



Cable colors: white = router ↔ box · purple = router ↔ reader · orange = router ↔ house internet (optional). Thick arrows = power; dotted = WiFi. The alarm light/siren plugs into the **Shelly Plug**, so the Shelly switches its power.

B. Turn it on / check it's alive — every time

1. On your **laptop**, join the Slate WiFi — network `TrakRF` , password `trakrf.id` .
2. Open **Chrome** → `https://app.demo.trakrf.id` , log in.
Page won't load? Re-check WiFi, wait a minute, reload. Still nothing → **call Mike**.
3. Open **Settings** → **Live feed**. Hold a tagged item near the doorway antenna — reads should pop up. **Reads = the reader is alive**.

TrakRF
Handheld Tag Reader
v1.2.0-353-gb09ce98a

Device Status
● **Disconnected**

- Home
- Inventory
- Locate
- Barcode
- Assets
- Locations
- Reports
- Mustering
- Settings
 - Readers
 - Live Feed**
 - Outputs
 - Geofence defaults
- Help

Demo Environment

Disconnected
M

Live Reads

Live tag inventory — every tag in range (registered or not) with read count and RSSI, for antenna & RSSI coverage tuning. A tag drops out after 30s of silence.

00:08
● Connected

☐ Split by antenna
☐ Show all BLE devices

EPC / MAC	READER	ANT	READS	LAST RSSI	AVG	MIN	MAX	AGE
D78304EBC99C	trakrf.id/mk107-23/reads	1	7	-49	-55	-62	-49	1s
300703506000000010005009	trakrf.id/cs463-20/reads	1	29	-43	-43	-45	-41	0s
300703506000000010005010	trakrf.id/cs463-20/reads	1	30	-43	-44	-47	-42	0s
300703506000000010005011	trakrf.id/cs463-20/reads	1	29	-54	-53	-60	-49	0s
E200001B120A013808307149	trakrf.id/cs463-20/reads	1	29	-49	-50	-54	-47	0s
A7B70000000000000014737	trakrf.id/cs463-20/reads	1	9	-63	-64	-67	-62	0s

TAGS IN VIEW
6

READERS
2

RSSI RANGE
-63 ... -43 dBm

READ RATE
—

Live reader feed — every tag read with age expiry, for antenna placement and RSSI coverage tuning

Settings → Live feed — tags appear here when the reader sees them.

- Open **Settings** → **Outputs**, expand the door strobe, click **Test-fire**. The strobe flashes; it clears itself after a few seconds (or click **Reset (off)**).

Type *

Shelly Gen4

Transport *

MQTT (broker)

Command Topic *

trakrf.id/alarm-plug

The Shelly's MQTT topic prefix. Must match the prefix configured on the device; the backend publishes to <topic>/command/switch:<id>. Firewall-friendly (no inbound).

Mode

Egress — fire on crossing, then latch

Egress alerts when an asset crosses. Presence stays on while an asset is here and clears it when the last one leaves.

RSSI threshold (dBm)

System default

Minimum signal strength for this output to react (stronger is closer to 0). Blank = org/system default.

Age-out (seconds)

System default

Re-arm window before the same tag can fire again.

Auto-off (seconds)

3

Device flips itself off after N seconds. 0 or blank = stay on until manual reset.

Cancel

Update Output Device

TEST

⚡ Test-fire

🔌 Reset (off)

Settings → Outputs → expand the strobe → **Test-fire / Reset (off)** at the bottom.

✓ **You're green when:** the page loads, Live feed shows reads, and Test-fire flashes the strobe. If all three work, the demo will work.

B2. Dial in the range (transmit power) — your tuning knob

Each antenna has a **transmit-power slider**. More power = reads tags from farther away; less power = only catches tags right at the door.

1. **Settings** → **Readers**, click your reader to expand it — a row per antenna, each with a **power slider (dBm)**.
2. Keep **Live feed** open in another tab while you test-walk.
3. Walk the tagged item through — including **in a bag / against your body**:
 - Doesn't catch the concealed item? **Slide power up.**
 - Catches tags from across the room / standing near? **Slide power down.**
4. Aim for: walking *through* always fires; standing *near* doesn't. It applies live (no save button).

ANTENNAS & LOCATION

csl_cs463

10–31.5 dBm

Read Timing

Dwell (ms)

Dedup window (ms)

☒ Antenna differentiation

Effective revisit cadence $\approx$ 500 ms (500 ms $\times$ 1 enabled)

Dedup window is below dwell — intra-dwell redundant publishing (the bottleneck).

Apply read timing

RSSI gate -80 dBm (read-only)

☒ 1 Doorway 1 (DOORWAY-1)

30.0 dBm

☐ 2 —set location—

30.0 dBm

☐ 3 —set location—

30.0 dBm

☐ 4 —set location—

30.0 dBm

Settings → Readers → expand the reader → one slider per antenna. Drag to tune.

This is the one setting you'll actively tune in the room. Everything else Mike pre-set. Can't find a setting that works? **Call Mike.**

C. Run the demo

- Calm state** — strobe off. *"Nothing tracked is leaving."*
- The catch** — walk the **registered** item through at a normal pace. **Strobe flashes within ~1 second.** *"That just walked out — caught."*
- Reset** — it clears itself (a few seconds), or Outputs → Reset.
- The point** — walk the **decoy** (not registered) item through. **Nothing happens.** *"Random tagged goods don't false-alarm — only your tracked assets do."*
- Real-world version** — carry the registered item in a **bag or against your body**. Catches most of the time. Honest line: *"~80–90% even concealed — and at replacement cost, catching most is the win."*

D. If something looks wrong (what you can do)

You see...	Try this
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Page won't load	Laptop on Slate WiFi? Reload. Wait 1 min. Still down → call Mike .
No reads in Live feed	Power-cycle the reader (unplug, replug, wait ~2 min).
Walking through doesn't fire	Reset the strobe, try again. Still nothing → call Mike (a settings tweak).
Strobe flashes for <i>everything</i>	Call Mike — needs a sensitivity tweak.
Strobe stuck ON	In the app: Outputs → Reset (off) .
Strobe never flashes, even on Test-fire	Check it's plugged in / powered. Still nothing → call Mike .
Box got unplugged / power blip	Plug back in, power on, wait ~2 min, redo section B .

Golden rules

- ✓ **You handle:** plugging things in, powering on, clicking in the app.
- 📞 **Call Mike for:** anything that needs a setting changed, or that a power-cycle + wait doesn't fix.
- ✗ **Never** run software updates or connect the box to venue/hotel internet during a demo — it's meant to run on its own.